

HK IMO Training 2020-2021
Problem Set 10
For 9 January 2020



Problem 37. Let $\triangle ABC$ be a scalene triangle. The median from A meets BC and the circumcircle of $\triangle ABC$ again at M and N respectively. The tangent at A to the circumcircle of $\triangle ABC$ meets the extension of BC at D . Let K be the point such that $ADMK$ is a parallelogram. Prove that $KA = KN$.

Problem 38. Let A_1 and C_1 be the feet of internal angle bisectors from A and C of $\triangle ABC$ respectively, and let I be the incentre of $\triangle ABC$. The circumcircle of $\triangle AIC_1$ meets the arc AC not containing B of the circumcircle of $\triangle ABC$ at C_2 . The circumcircle of $\triangle CIA_1$ meets the arc BC not containing A of the circumcircle of $\triangle ABC$ at A_2 . Prove that the lines A_1A_2 and C_1C_2 intersect on the circumcircle of $\triangle ABC$.

Problem 39. There is a sphere in the space. Let A, B, C, D be four points in the space such that each of AB, BC, CD, DA is tangent to the sphere. Prove that all tangential points are coplanar.

Problem 40. A tetrahedron $ABCD$ is given. The incircles of $\triangle ABC$ and $\triangle ABD$ with centres O_1 and O_2 touch AB at points T_1 and T_2 respectively. Let Π_{AB} be the plane passing through the midpoint of T_1T_2 and is perpendicular to O_1O_2 . Define the planes $\Pi_{AC}, \Pi_{BC}, \Pi_{AD}, \Pi_{BD}, \Pi_{CD}$ analogously. Prove that these 6 planes have a common point.